Lecture Notes on

Celestial Motion and Orbits

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These notes summarize the second Astrophysics Club meeting, where we introduced gravity, orbital motion, Kepler's laws, and the expansion of the universe.

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1 Why Celestial Motion Matters

Understanding celestial motion allows us to:

- Predict eclipses and planetary alignments.
- Determine orbital periods (e.g. Earth's 365-day revolution).
- Measure stellar velocities using Doppler shifts.
- Model long-term evolution of planetary systems.

Celestial mechanics connects observation directly to physics.

2 Historical Development of Orbital Models

- **Ancient Times:** Geocentric model (Earth-centered universe).
- **1543 – Copernicus:** Heliocentric model.
- **1609 – Kepler:** Empirical laws of planetary motion.
- **1687 – Newton:** Universal law of gravitation.
- **1915 – Einstein:** General relativity.

Each step improved predictive power and physical understanding.

3 Free Fall and Orbits

Important conceptual point:

- Objects in orbit are in **continuous free fall**.
- Free fall means gravity is the only force acting.
- The International Space Station is in free fall.

Orbit = falling sideways fast enough to miss the Earth.

4 Newton's View of Gravity

Newton proposed:

- Gravity is a force.
- It acts between any two masses.
- It acts instantaneously (classically).

The gravitational force is:

$$F = \frac{Gm_1m_2}{r^2}.$$

Where:

- G is the gravitational constant.
- r is the distance between objects.
- Force decreases with $1/r^2$.

4.1 Orbital Speed Derivation

For circular orbit:

Gravitational force provides centripetal force:

$$\frac{GMm}{r^2} = \frac{mv^2}{r}.$$

Cancel m:

$$v = \sqrt{\frac{GM}{r}}.$$

This shows:

- Closer planets move faster.
- Orbital speed depends only on central mass and radius.

5 Einstein's View of Gravity

Einstein's theory of general relativity states:

- Gravity is not a force.
- Mass and energy curve spacetime.
- Objects follow straight paths (geodesics) in curved spacetime.

Consequences:

- Gravitational lensing.
- Time dilation.
- Black holes.

Newton's theory works extremely well for most planetary motion, but Einstein's theory is more complete.

6 Kepler's Laws of Planetary Motion

6.1 First Law: Elliptical Orbits

- Planets orbit in ellipses.
- The Sun is at one focus.
- Perfect circular orbits are rare.

An ellipse has:

- Semi-major axis a.
- Two foci.

6.2 Second Law: Equal Areas

- A line from the Sun to the planet sweeps equal areas in equal times.
- Planets move faster when closer to the Sun.
- Planets move slower when farther away.

This follows from conservation of angular momentum.

6.3 Third Law: Period–Radius Relationship

Kepler discovered:

$$T^2 \propto a^3.$$

Using Newton's law:

$$\frac{GMm}{r^2} = \frac{mv^2}{r}, \quad v = \frac{2\pi r}{T}.$$

Substitute:

$$T^2 = \frac{4\pi^2}{GM} r^3.$$

Thus:

- Farther planets have longer orbital periods.
- The constant depends only on the central mass.

7 Expansion of the Universe

The universe is expanding.

Key points:

- Expansion began with the Big Bang.
- Gravity attempts to slow expansion.
- Observations show expansion is accelerating.
- Acceleration is attributed to dark energy.

Dark energy behaves like a negative pressure that drives expansion.

8 Hubble's Law

Edwin Hubble observed:

$$v = H_0 d.$$

Where:

- v = recession velocity.
- d = distance to galaxy.
- $H_0 \approx 70\,\mathrm{km/s/Mpc}$.

Implications:

- The farther a galaxy is, the faster it recedes.
- Space itself is expanding.

Rough age estimate of universe:

$$t \sim \frac{1}{H_0}.$$

Which gives about 14 billion years.

9 Looking Ahead

Future topics include:

- Relativity
- Black holes
- Stellar evolution
- Cosmology

Celestial motion forms the foundation for all advanced astrophysics.